CLASSIFICATION OF ROCKS AND ROCK CYCLE

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- The earth is composed of various kinds of elements.
- About 98% of the total crust is made up of eight elements as oxygen, silicon, aluminium, iron, calcium, sodium, potassium, and magnesium.
- The rest is constituted by elements like titanium, hydrogen, phosphorous, manganese, Sulphur, carbon, nickel and others.
- The elements in the earth's crust are rarely found exclusive but are usually combined with other elements to make various substances.
- These substances are recognised as minerals.

The table below shows the percentage wise share of each element in both the earth's crust and the whole earth.

Earth's Crust	Entire Earth
Oxygen-47%	Iron-35%
Silicon-28%	Oxygen-30%
Aluminium-8%	Silicon-15%
Iron-5%	Magnesium-13%
Magnesium-4%	Nickel-2%
Calcium-2%	Sulphur-2%
Pottassium-2%	Calcium-1%
Sodium-2%	Aluminium-1%
Others-2%	Others-1%

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MINERALS IN THE EARTH'S CRUST

- A mineral is a naturally occurring organic or inorganic substance, having an orderly atomic structure and a definite chemical composition and physical properties.
- A mineral is composed of two or more elements. But, sometimes single element minerals like sulphur, copper, silver, gold, graphite, etc are also found.
- The basic source of all minerals is the hot magma in the interior of the earth.
- When magma cools, crystals of the minerals appear and a systematic series of minerals are formed in sequence to solidify so as to form rocks.

- The minerals which contain metals are called as metallic minerals (eg: Haematite) and the metallic minerals which are profitably mined are called as the ores.
- The crust of the earth is made up of more than 2000 minerals, but out of these, only six are the most abundant and contribute the maximum.
- These six most abundant minerals are feldspar, quartz, pyroxenes, amphiboles, mica and olivine.

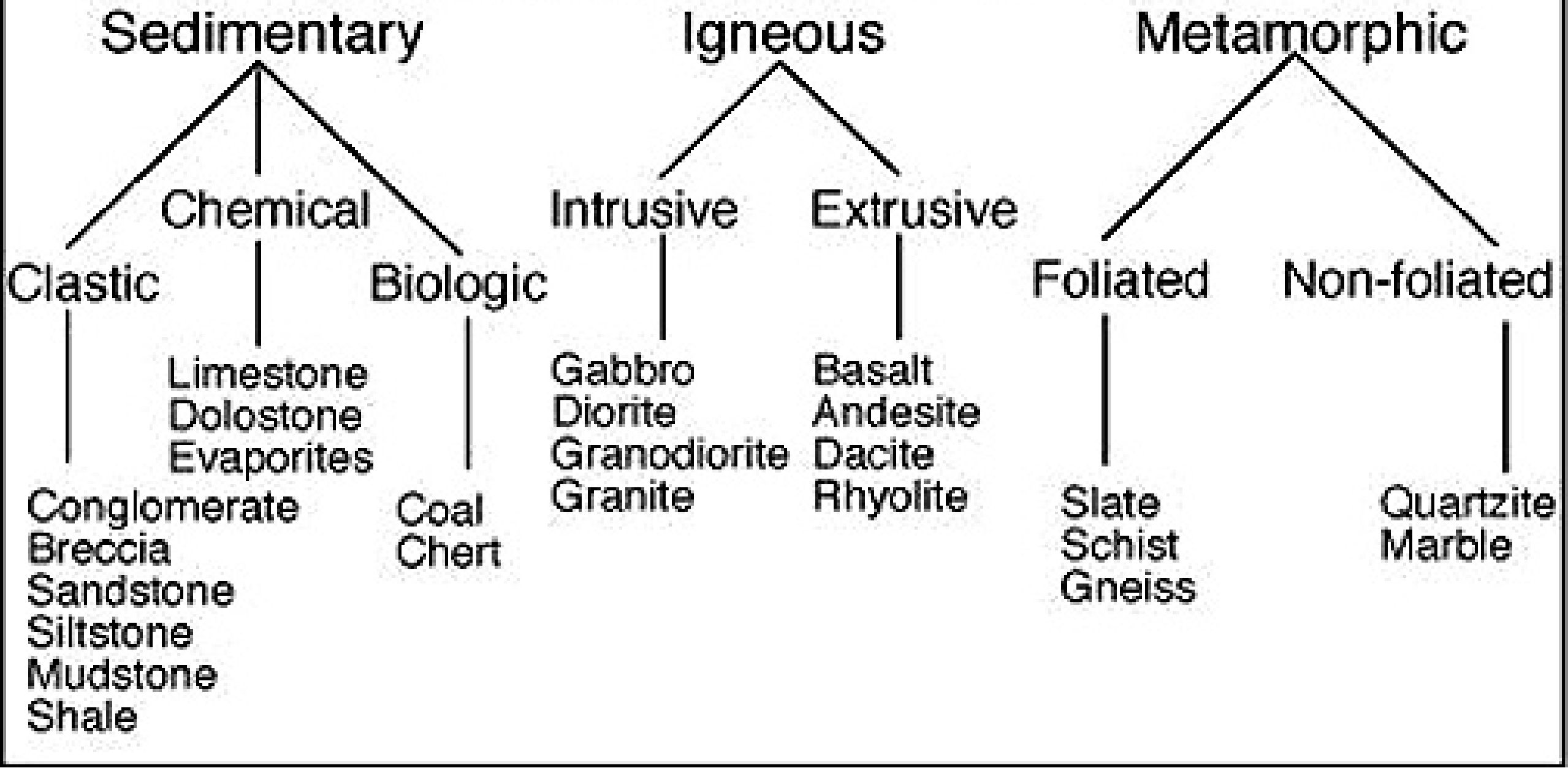
AMPHIBOLES



ROCKS IN THE EARTH'S CRUST

- A rock is but a composition of minerals.
- They are aggregates or a physical mixture of one or more minerals.
- Rocks may be hard or soft and in varied colours.
- Feldspar and quartz are the most common minerals found in all type of rocks.
- The science dealing with the study of rocks is called as Petrology.

CLASSIFICATION OF ROCKS



Sedimentary Rocks are classified based on their components.

Clastic



1.) Conglomerate



2.) Sandstone



3.) Shale

Organic



1.) Coal



2.) Fossiliferous Limestone

Chemical

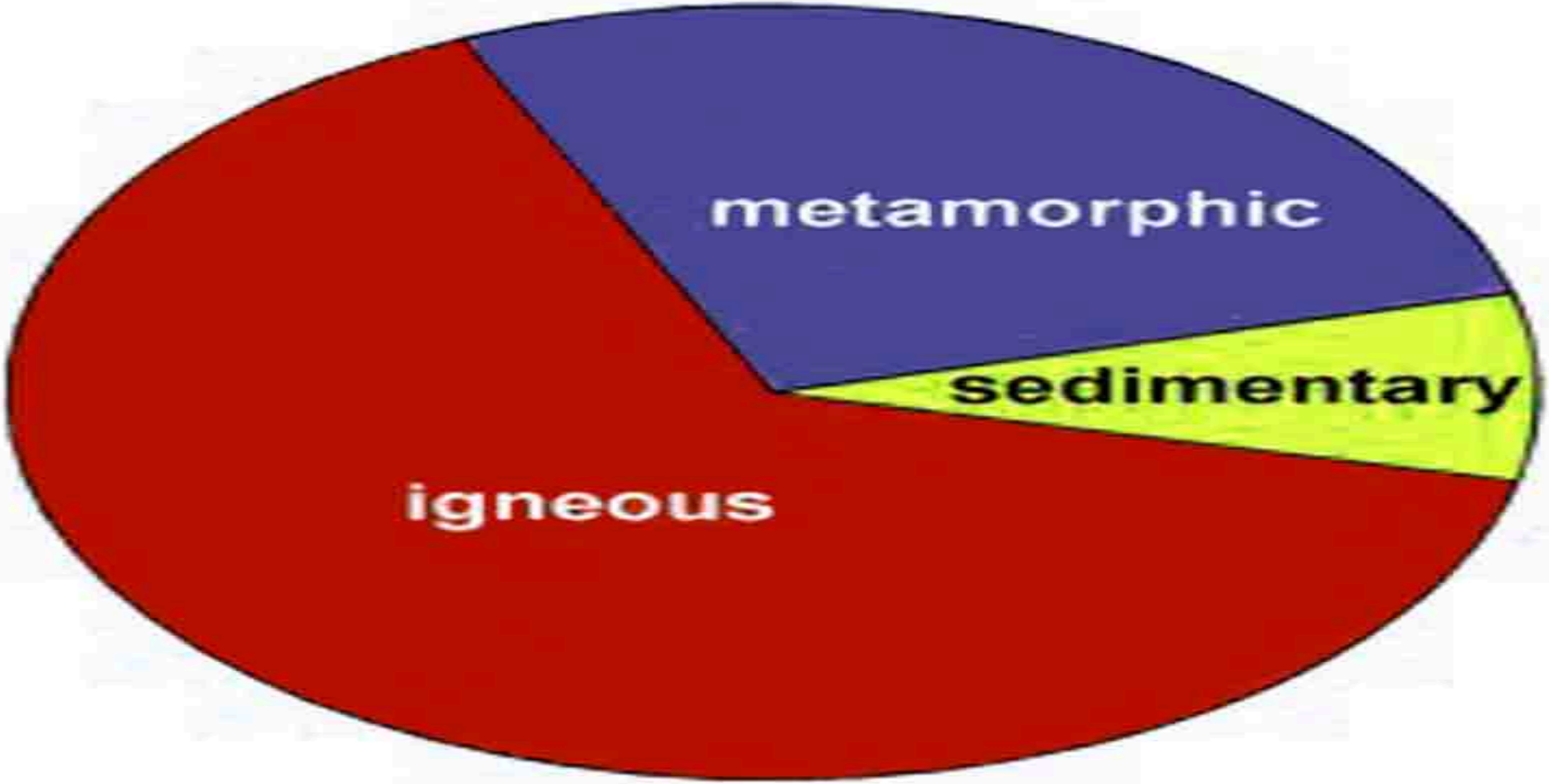


1.) Limestone



2.) Salt

- On the basis of mode of formation, rocks may be classified into three:
- Igneous Rocks
- Sedimentary Rocks
- Metamorphic Rocks


metamorphic
sedimentary
igneous

IGNEOUS ROCKS

- Igneous rocks are formed by the cooling of highly heated molten fluid material called as Magma.
- Asthenosphere, which is just below the upper mantle, a region beneath Lithosphere is the main source of magma.
- They might be formed directly by cooling of magma from the interior of the earth itself or by cooling of lava from the surface of the earth.

- As they comprise the earth's first crust and all other rocks are derived from them, they are also called as the parents of all rocks or the Primary Rocks.

- They are the most abundant rocks in the earth's crust.

- On the basis of their mode of occurrence, igneous rocks can be classified as Intrusive and Extrusive Igneous Rocks.

SEDIMENTARY ROCKS

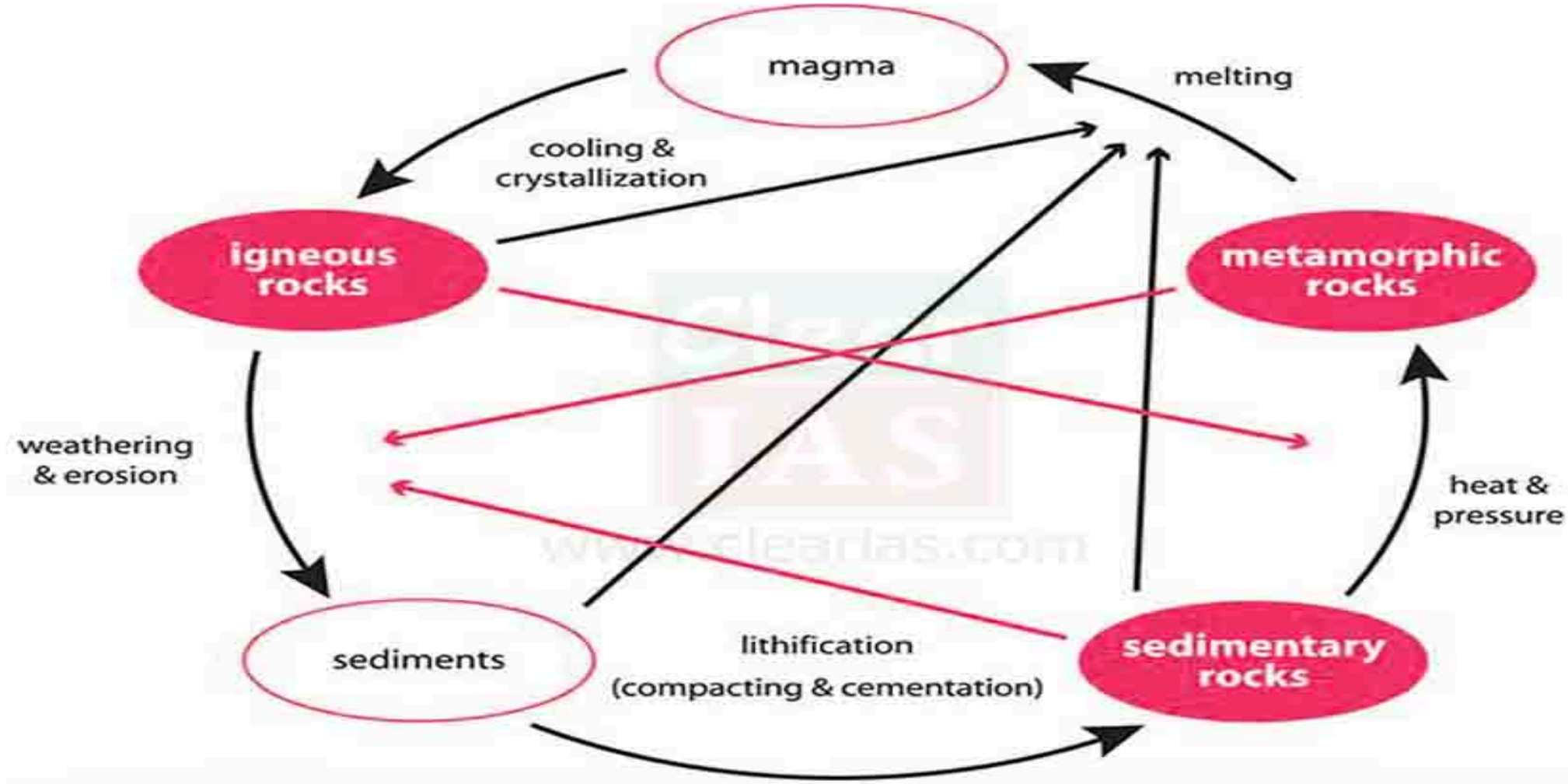
- These rocks are formed by successive deposition of sediments.
- These sediments may be the debris eroded from any previous existing rock which may be igneous, metamorphic or old sedimentary rocks.
- The process of successive deposition and formation of sedimentary rocks is called as Lithification.
- Due to successive depositions, they have a layered or stratified structure and hence are also called as Stratified Rocks.

METAMORPHIC ROCKS

- The word metamorphic means ‘change of form’.
- Hence, these rocks form under the action of temperature, pressure and volume changes on original rocks.
- Metamorphic rocks are formed under the influence of heat or pressure on original rocks which cause to change their colour, hardness, structure and composition.
- The process of recrystallization and reorganisation of materials within the original rock is called as metamorphism.

- When the metamorphism happens without any appreciable chemical change, it is called as Dynamic Metamorphism.
- If metamorphism happened due to the influence of heat, it is called as Thermal Metamorphism. It has two types: Contact Metamorphism and Regional Metamorphism.
- When the reorganisation occurs due to direct contact with the hot magma, it is called as Contact Metamorphism.
- If the rocks undergo reorganisation due to tremendous heat/ pressure formed as a result of tectonic shearing, it is called as Regional Metamorphism.
- Metamorphic Rocks can be classified into Foliated (Slate, Schist, Gneiss) and Non-Foliated (Quartzite, Marble) Metamorphic Rocks on the basis of the presence or absence of bands of mineral grains.

ROCK CYCLE


magma
melting
cooling &
crystallization
igneous
rocks
metamorphic
rocks
weathering
& erosion
heat &
pressure
sediments
lithification
(compacting & cementation)
sedimentary
rocks